

REVIEW

Antibacterial Mechanisms of Two-dimensional Nanomaterials

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Manuscript received: November 7, 2019

Keywords: antibacterial, mechanisms, antimicrobial, two dimensional, nanomaterials

1. Introduction:

Multi-drug resistant bacterial infection is the critical global challenge. According to the United States *Food and Drug Administration* (FDA), every year around 48 million cases of bacterial foodborne illness are estimated in the U.S. itself among which approximately 128,000 cases are of hospitalizations and 3000 cases are associated with deaths.¹ Multi-drug resistant and constantly emerging Methicillin-resistant *Staphylococcus aureus* (MRSA) gram-positive bacteria and *Shigella* spp. (sister-species of Genus *Escherichia*) rod shaped gram-negative bacteria are within WHO top priority list for which new antibiotics are needed on the urgent basis.²⁻⁴ *E. coli* bacteria lives in human intestine and their most of the types are harmless, but *E. coli* O157:H7 gram-negative specific strain produces Shiga toxins which can cause kidney failure by damaging the red blood cells.⁵ Likewise, multiple drug resistant gram-positive MRSA bacteria are responsible for sepsis infections which can affect lungs, skin, gut, urinary tract and kidney.^{6,7} Antibacterial resistance evolves with misuse or overuse of drugs.⁸ Bacterial antibiotics resistance biochemical mechanisms can be categorized as enzymatic inactivation of antibiotic drug molecule, drug efflux pumps made-up with chromosome and/or plasmid, altering the intracellular target of drug by involving ribosomes, metabolic enzymes, proteins for cell wall synthesis and DNA replication and chromosomal transfer by acquiring DNA from another bacteria.⁹⁻¹¹ Over the period, several studies have been performed to investigate toxins secretion function and the antibiotic resistance mechanisms of MRSA superbugs.^{12,13} To combat against continuously emerging anti-biotics resistance, there is a need to develop novel platforms to achieve synergistic and rapid bacterial killing. The bacterial antibiotics resistance various biochemical mechanisms are represented in the **Figure 1**.

Organization	18(+)	
Introduction	20	
Expt. Setup/ Analysis	18(+)	
Conclusion	18(+)	
Total	74	77

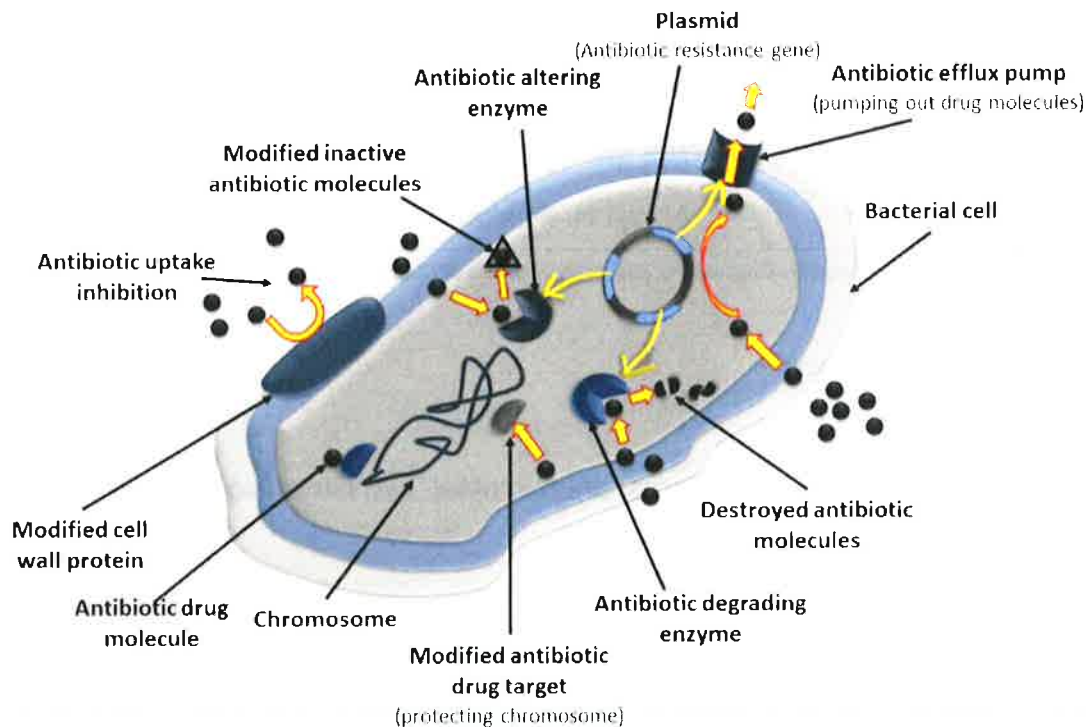


Figure 1. The bacterial antibiotics resistance biochemical mechanisms

Development of bactericidal nanomaterial is an appealing field in science which has tremendous capacity to address superbugs infections issues. The nanoparticles are being used as an antibacterial agent considering their advantages of extremely high surface area to volume ratios and respective changes in their mechanical, chemical, electrical, optical and magnetic properties.¹⁴ Among various nanoparticles such as copper oxide (CuO), nickel oxide (NiO), zinc oxide (ZnO), antimony trioxide (Sb₂O₃), CuO NPs are considered to be more toxic considering the toxicity of ions produced from the nanoparticles which apparently depend on the natural toxic properties of heavy metals.¹⁵ Oxidation of NPs in contact with aqueous media initiate the release of the metal ions.¹⁶

Among the family of advanced nanomaterials, two-dimensional (2D) nanomaterials have shown tremendous potential considering their benefits over very high surface to volume aspect ratio as compared with nanoparticles and numerous surface active sites.^{17,18} Herein, we provide brief overview of different antibacterial mechanisms of 2D nanomaterials which will provide further insights to extend their applications in biomedical field. In the family of 2D nanomaterials, the antibacterial activities of reduced graphene oxide (rGO) and graphene oxide (GO) nanocomposites are widely studied for their physical damage abilities against bacterial cell membranes and their capacity to produce reactive oxygen species (ROS), ascribed to the presence of surface functional groups and the large effective surface area.^{19,20} Antibacterial activities of vertically aligned molybdenum disulfide (MoS₂) and manganese dioxide (MnO₂) 2D nanosheets show possible antibacterial mechanisms mainly associated with cell membrane oxidative damage with ROS and physical penetration into bacterial cell membrane.^{17,18} Lately, researchers have focused their attention towards the phosphorene i.e. 2D exfoliated black phosphorus (BP)

nanosheets associated with their unique attributes, such as ease of synthesis, high anisotropic charge carrier mobility, singlet oxygen generation capacity and nanosheets thickness dependent tunable intrinsic bandgap which can be adjusted between zero bandgap graphene and wide bandgap transition metal dichalcogenides.²¹⁻²⁵ The following section includes the brief overview of various possible antibacterial mechanisms associated with 2D nanomaterials.

2. Antibacterial mechanisms of 2D nanomaterials:

The cell envelop of *E. Coli* Gram-negative bacteria comprises of three layers, such as, cytoplasmic inner membrane layer comprising of lipid-proteins, outer membrane containing lipopolysaccharides and proteins and intermediate thin (~7 nm thickness) rigid peptidoglycan mesh located in the periplasmic space.²⁶ Peptidoglycan layer comprising of polymeric chains of N-acetylmuramic acid and N-acetylglucosamine linked with peptides is responsible for the rigidity and the cell membrane integrity.^{27,28} The cell wall structural differences among gram-negative and gram-positive bacteria are represented in the **Figure 2**.

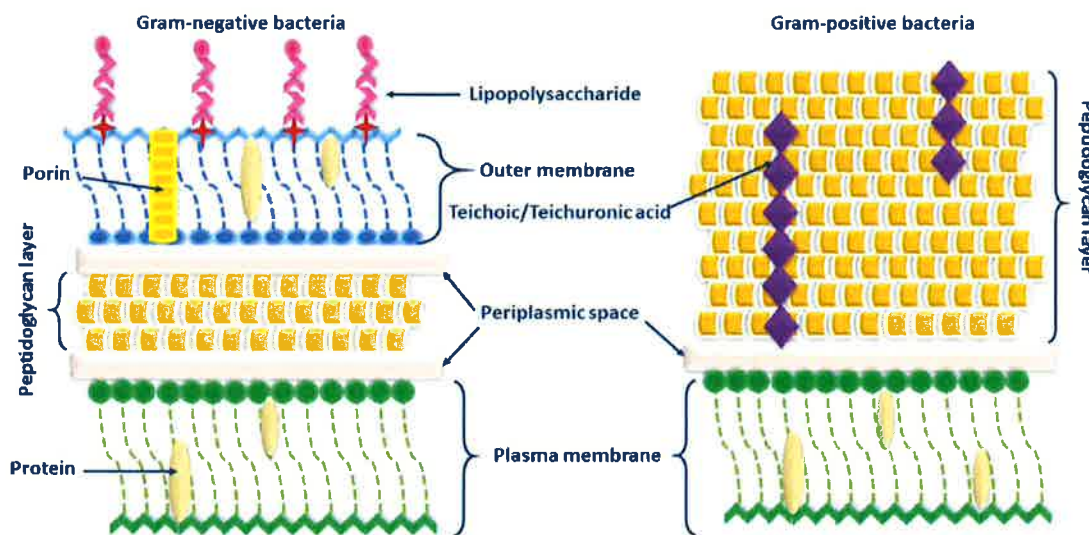


Figure 2. The gram-positive and gram-negative bacterial cell wall structure.

2D nanomaterials upon physical contact with both gram-positive or gram-negative bacterial cell membrane can cause membrane damage by ROS dependent or ROS independent pathways.²⁹ Upon UV irradiation ultrathin few-layer 2D nanosheets possess high anisotropic hole mobility and tunable semiconductor bandgap, which can mainly generate singlet oxygen and superoxide radical as exogenous ROS.³⁰⁻³⁴ Among the family of ROS such as singlet oxygen ($^1\text{O}_2$), superoxide radical (O_2^-), hydrogen peroxide (H_2O_2) and hydroxyl radicals ($^*\text{OH}$), singlet oxygen is more dominant, as it possesses longer lifetime.³⁵ ROS can generate oxidative stress in addition with reacting with cell essential macromolecules such as DNA, RNA, proteins and damaging cell membrane and

proteins.³⁴ ROS can also oxidize lipid cell membrane and hence can rupture the bacterial membranes.³⁶ The oxidative stress inside cell membrane can also be generated by charge transfer occurred between 2D nanosheet and bacterial cell membrane, which eventually interrupts the electron transport in the membrane respiratory chain leading towards the depletion of intracellular adenosine triphosphate (ATP) levels.³⁷ Depletion of ATP levels in *E. Coli* bacteria disrupts metabolic pathways, which can promote the production of endogenous ROS in the form of O_2^- and H_2O_2 .³⁸ Additionally, certain functionalized 2D nanomaterials are capable of inducing acidic cell environment in the presence of water.^{39,40} pH homeostasis (maintaining cytoplasmic pH by controlling transmembrane pH) is very critical function for bacteria as proton concentration, as proton motive force is essential for cellular bioenergetics.⁴¹ Neutralophilic bacteria maintain cytoplasmic pH within very narrow window as 7.5-7.7 pH, even in wide range of external environment pH as 5-9 pH.^{41,42} Studies confirm that in *E. Coli* bacteria biofilm, cells shifted from 7.5 to 5.5 pH cannot recover cytoplasmic pH, which apparently affects the protein synthesis and bioenergetic mechanisms in bacteria.^{41,43} Acids such as phosphoric acid and sulfuric acid are responsible for killing bacteria mainly due to increased hydrogen ions concentration causing acid stress, as it adversely affects transmembrane proton gradient.^{43,44} Additionally, the wrapped flexible 2D nanosheets can isolate bacteria from surrounding nutrients and hence confine the cells proliferation.^{45,46} The penetration of 2D nanomaterials in the bacterial cell membrane can be attributed with their sharp edges. The physical penetration of 2D nanosheet can damage the cell wall by triggering the intracellular periplasmic and cytoplasmic leakage. The nano-knives like behavior of 2D nanomaterials penetrating the cell membrane are noted by Pham et al.⁴⁷ for graphene nanosheets. The various possible antibacterial mechanisms are represented in **Figure 3**.

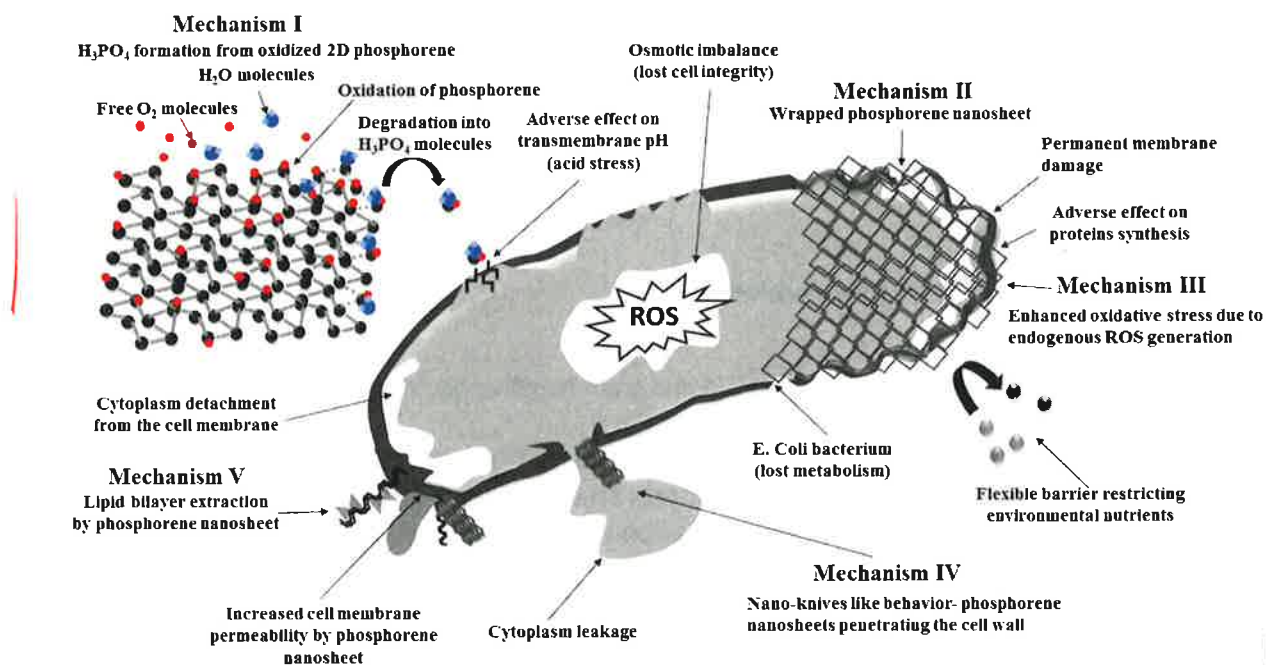


Figure 3. Representative schematic of possible antibacterial mechanisms of 2D nanomaterials

3. Summary and outlook:

This review summarizes various antibacterial mechanisms of 2D nanomaterials. As compared with metal nanoparticles, 2D nanomaterials have exhibited significantly enhanced antibacterial activities affiliated with their extremely high surface to volume aspect ratio, plenty of surface active sites and controlled toxicity.^{17,18} The antibacterial properties of 2D nanomaterials are believed to be caused through chemical and physical pathways upon having physical contact with bacteria. To extend the applications of 2D nanomaterials as antimicrobial agents, it is important to understand their structural physiochemical characteristics such as thickness, size, synthesis method and oxidation capacity of synthesized nanomaterials.

The understanding of antibacterial mechanisms of 2D nanomaterials is helpful in expanding their scope in the biomedical field and in developing more sophisticated systems for gene delivery, drug delivery, biosensing, bioimaging, cancer treatment, arthroplasty surgeries.⁴⁸ On the other hand, in vivo and in vitro studies of 2D nanomaterials have demonstrated excellent cytocompatibility even at high concentrations with very negligible changes in the cellular morphology.⁴⁹⁻⁵²

Moreover, the contamination of pathogenic microorganisms in water is a worldwide concerning issue, which ask for economical and efficient safe drinking water filtration systems.⁵³ Functionalization capabilities of 2D nanomaterials make them ideal for controlling hydrophobic-hydrophilic characteristics of the interface at the filtration membrane materials. Conclusively, 2D nanomaterials can exhibit excellent antimicrobial properties at various fronts to combat against rising superbugs issues as they can carry synergistic efficient killing.

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